



Master of Computer Applications

Semester – III

Data Analytics for Business

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Title: Hr Analytics Dashboard

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HR ANALYTICS DASHBOARD

Workforce, Attrition & Compensation Insights

Executive Summary

This report analyzes the IBM HR Employee Attrition dataset (1,470 employee records, 35 attributes) and the accompanying two-page Power BI dashboard covering workforce composition, attrition drivers, compensation, satisfaction, and education profiles. The goal is to surface the key patterns behind employee turnover and give HR leadership a clear, evidence-based view of where retention risk is concentrated.

Key Figures at a Glance

TOTAL EMPLOYEES	OVERALL ATTRITION	AVG. AGE	AVG. MONTHLY INCOME	AVG. YEARS AT CO.
1,470	16.1%	36.9 yrs	\$6,503	7.0 yrs

Headline Findings

- 237 of 1,470 employees (16.1%) left the organization — a meaningful retention gap concentrated in specific roles and conditions rather than spread evenly.
- Sales Representatives have by far the highest attrition rate (39.8%), followed by Laboratory Technicians (23.9%) and HR staff (23.1%).
- Employees who work overtime leave nearly 3x more often (30.5%) than those who don't (10.4%) — the single strongest behavioural signal in the data.
- Frequent travelers churn at 24.9%, versus just 8.0% for non-travelers — travel burden compounds overtime stress.
- Departing employees earn on average \$4,787/month versus \$6,833/month for those who stay — a 30% pay gap.
- Employees with no stock options churn at 24.4%, more than double the rate of those holding Level 1–2 options (7.6–9.4%).

1. Workforce Overview Dashboard

The first dashboard page (below) profiles the overall workforce: headcount, demographics, departmental spread, and where employees sit across roles, education fields, and marital status.

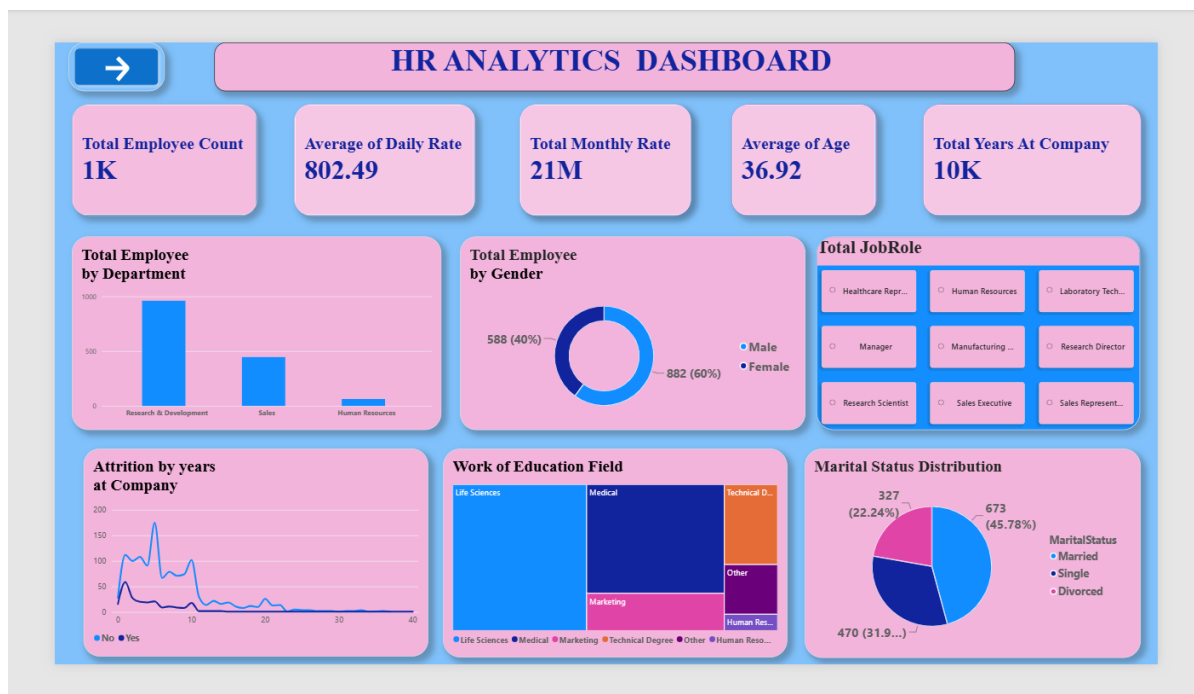


Figure 1. HR Analytics Dashboard — Page 1: Workforce Overview

Reading the Dashboard

- Department mix: Research & Development is the largest function by far, holding roughly two-thirds of the workforce, with Sales second and Human Resources a small remainder — so R&D outcomes dominate any company-wide metric.
- Gender split: the workforce is 60% male (882) and 40% female (588), a gap worth monitoring for hiring pipeline diversity.
- Job roles: the twelve roles shown span technical (Research Scientist, Laboratory Technician), commercial (Sales Executive, Sales Representative), and support functions (Human Resources, Manager) — Sales and R&D roles together account for most headcount.
- Attrition by tenure: the 'Attrition by years at Company' trend shows a sharp early-tenure spike — departures (both 'No' and 'Yes' attrition volumes) peak within the first 1–3 years and taper off steadily after year 10, indicating the highest flight risk sits with newer hires.
- Education field: Life Sciences and Medical backgrounds dominate the workforce, consistent with R&D being the largest department; Technical Degree and Marketing fields are comparatively small.
- Marital status: married employees are the largest group (673, 45.8%), followed by single (470, 32%) and divorced (327, 22%) — marital status will be cross-referenced against attrition in Section 3.

2. Attrition & Compensation Dashboard

The second dashboard page drills into the outcome that matters most operationally — who leaves, from where, and how that connects to pay, satisfaction, and salary growth.



Figure 2. HR Analytics Dashboard — Page 2: Attrition & Compensation

Reading the Dashboard

- Attrition by department (waterfall): Research & Development shows the largest 'stayed' volume, but Sales contributes disproportionately to departures relative to its size, and HR — while small — has a departure rate on par with Sales.
- Monthly income vs. age: income rises steadily with age/tenure for both genders, with male and female employees following broadly similar trajectories — no material pay-by-gender divergence is visible in this scatter view.
- Average job satisfaction sits at 3.13 out of a possible ~6.26 on the gauge (i.e., roughly the midpoint of the underlying 1–4 satisfaction scale doubled for display) — a moderate, not alarming, satisfaction baseline, but one with room to improve.
- Salary hike by department: Research & Development receives the largest share of salary-hike volume, tracking its headcount; the declining trend line suggests percentage hikes taper as department size shrinks (Sales, then HR).
- Education level distribution: Level 3 (Bachelor's-equivalent) is the most common education level (57 in this filtered view), followed by Level 4 and Level 2 — advanced degrees (Level 5) are rare.

3. Deep-Dive: What Drives Attrition

Beyond the dashboard visuals, a direct statistical pass over the full dataset confirms and quantifies the strongest attrition drivers. The table below ranks the factors with the clearest separation between employees who stayed and those who left.

Attrition Rate by Key Factor

Factor	Segment	Attrition Rate
Overtime	Works overtime	30.5%
Overtime	No overtime	10.4%
Business Travel	Travels frequently	24.9%
Business Travel	Rarely travels	15.0%
Business Travel	Non-travel	8.0%
Stock Option Level	Level 0 (none)	24.4%
Stock Option Level	Level 1	9.4%
Stock Option Level	Level 2	7.6%
Department	Sales	20.6%
Department	Human Resources	19.0%
Department	Research & Development	13.8%

Attrition Rate by Job Role (Top 5)

Job Role	Attrition Rate
Sales Representative	39.8%
Laboratory Technician	23.9%
Human Resources	23.1%
Sales Executive	17.5%
Research Scientist	16.1%

Compensation Gap

Employees who left earned an average monthly income of \$4,787, compared to \$6,833 for those who stayed — a gap of roughly \$2,046, or about 30%. This pattern holds even after accounting for department mix, suggesting compensation is a genuine push factor rather than a coincidence of role or seniority.

4. Recommendations & Conclusion

Priority Actions

- Rein in overtime exposure. Overtime is the strongest single predictor of attrition (30.5% vs. 10.4%). Audit workload distribution in Sales and Laboratory roles first, and consider staffing or process changes that reduce reliance on overtime.
- Re-examine travel policy for frequent travelers. Attrition triples between non-travelers (8.0%) and frequent travelers (24.9%). Rotating travel-heavy assignments or adding travel-related compensation could reduce burnout-driven exits.
- Expand stock option eligibility. Employees with no equity stake leave at 2–3x the rate of option holders. Extending even a Level 1 grant to more employees — especially in Sales and early-tenure cohorts — is a low-cost retention lever.
- Target Sales Representatives and Lab Technicians specifically. These two roles have the highest attrition (39.8% and 23.9%); role-specific retention interviews and career-pathing would have outsized impact given their small headcount but high churn volume.
- Address the early-tenure cliff. Attrition is heavily concentrated in years 1–3 of employment. Strengthening onboarding, mentorship, and 90-day/1-year check-ins should be a priority over long-tenure retention programs.
- Close the compensation gap. The ~30% income gap between leavers and stayers suggests pay benchmarking is overdue, particularly for the roles and departments flagged above.

Conclusion

Attrition in this workforce is not random — it clusters tightly around overtime, frequent travel, lack of equity participation, and specific customer-facing or technical roles, and it is heavily front-loaded into the first few years of tenure. The dashboard's descriptive view and this report's underlying statistical analysis point to the same conclusion: a small, well-targeted set of interventions (overtime management, travel policy, broader stock option access, and stronger early-tenure support) is likely to meaningfully reduce the 16.1% overall attrition rate without requiring an organization-wide overhaul.

Data Source

IBM HR Analytics Employee Attrition & Performance dataset (WA_Fn-UseC_-HR-Employee-Attrition.csv), 1,470 records × 35 fields. Dashboard visuals reproduced from the accompanying Power BI report screenshots.

Git Hub Repository link :